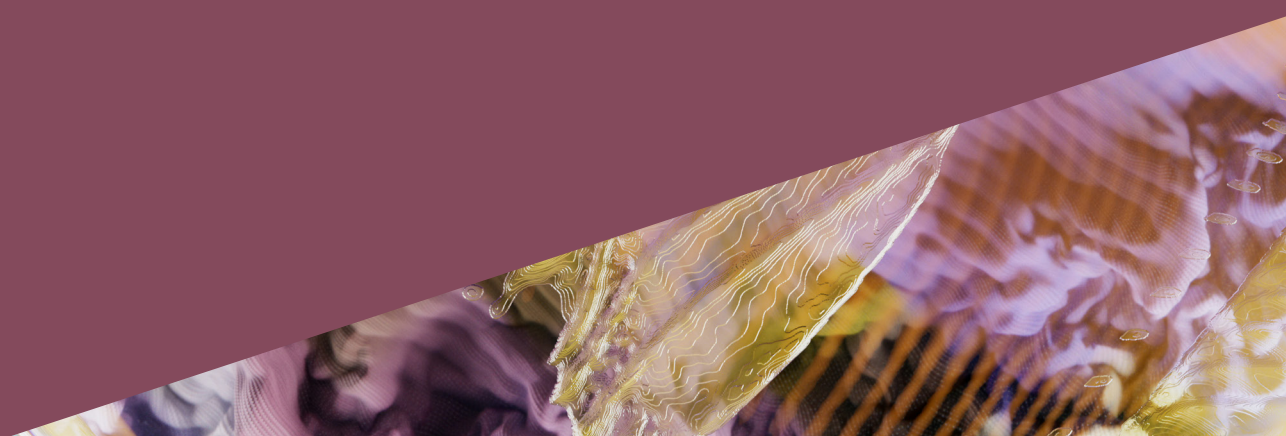


Inclusive and Secure Artificial Intelligence

*A Global Perspective on Policy
and Technical Developments*

Saiph Savage and Lili Savage



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Acknowledgements

We would like to extend our heartfelt thanks to all the participants of the March 2024 workshop for their valuable insights and generous contributions. Your expertise and thoughtful engagement greatly enriched the discussions and helped shape the direction of this report. In particular, we are grateful to:

Soma Marton, Anna Lagos, Urbano Hidalgo, Luis Cervantes, Hunjun Shin, Wanda Munoz, Meriem, Genwa Hassan, Valeria Tafoya, Shyam Krishna, Samantha Torres, Kathryn O'Neil, Xintong Zhang, Samyak Bijal Shah, Genoveva Vargas-Solar, Kasif Imteyaz, Hwi Joon Lee, Frida Ruh, Chirag Malhotra, Daniela Rojas, Angelica Fernandez, Zinnya del Villar, Rishabh Kumar, Haoran Xu, Mariel Garcia-Montes, Andy Yu, Jose Montano, Anahiby Becerril, Vedansh Kapoor, Manasa Rao, Renlei Huang, Can He, Karin Quinones, Sumanth Kumar Mohapatra, Nayyara Rahman, Fabian Christopher, Karen Selavy, Suchita Sharma, Hiranmai Devarasetty, Selma Rojas, Xinjie Shen, Punith Subashchandra, Akshita Singh, Ashley Tom, Nithin S., Madhumita Kolukuluri, Princewill Umeh, Rafael Morales, Marianne Teixido, Elena Estavillo, Sumaia Tambal, Debora de C. Leal, Ivana Putri, Sarah Widmaier, Justin Nappi, Dhruv Prakash Belai, Wenquin Wu, Kausha Trivedi, Miro Leon Bucher, Aditya, Asia Biega.

Thank you for your commitment to building more inclusive, secure, and culturally aware AI systems.

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Abbreviations

AI	Artificial Intelligence
AI4D	Artificial Intelligence for Development
CAC	Cyberspace Administration of China
CHI	Conference on Human Factors in Computing Systems
DSSG	Data Science for Social Good
EMNLP	Empirical Methods in Natural Language Processing
FAccT	Fairness, Accountability, and Transparency (also FAT*)
GPAI	Global Partnership on Artificial Intelligence
HITL	Human-in-the-loop
ICR	International Cultural Relations
ICT	Information and Communications Technology
LXAI	LatinX in AI
ML	Machine Learning
NGO	Non-Governmental Organization
NLP	Natural Language Processing
OGBV	Online Gender-Based Violence
SMR	Small Modular Reactor
TFGBV	Technology-Facilitated Gender-Based Violence
UN	United Nations
UNESCO	United Nations Educational, Scientific and Cultural Organization
UX	User Experience

Concepts

1. Artificial Intelligence (AI)

Artificial Intelligence (AI) refers to systems designed to perform tasks that require human-like intelligence—such as learning, reasoning, problem-solving, and language understanding. AI is not a single technology but an umbrella for various computational approaches that allow machines to simulate intelligent behavior.

2. Machine Learning (ML)

Machine Learning is a subset of AI where systems improve their performance by learning from data. Rather than relying on fixed rules, ML algorithms detect patterns and adapt over time. Supervised, unsupervised, and reinforcement learning represent different ways these systems are trained to make decisions or predictions.

3. Generative AI

Generative AI enables machines to create new content—text, images, audio, and code—based on patterns in existing data. Tools like ChatGPT and DALL-E exemplify this approach, opening possibilities for automation in creative industries, while also raising questions around originality, authorship, and bias.

4. Neural Networks and Deep Learning

Neural networks mimic the structure of the human brain, using interconnected layers of artificial neurons to process data. Deep learning, a branch of neural networks with many layers, powers tasks such as language translation and image recognition. Transformers, a specific architecture, underpin today's leading AI models.

5. Reinforcement Learning

Reinforcement Learning is a technique where AI agents learn optimal actions by interacting with an environment and receiving feedback in the form of rewards. This trial-and-error approach is effective in dynamic settings like robotics or recommendation systems, where behavior adapts in real time.

6. Explainable AI (XAI)

Explainable AI focuses on making algorithmic decisions transparent and interpretable. As AI systems are increasingly used in areas with real-world impact—like healthcare or finance—XAI enables users to understand the rationale behind predictions and builds trust through clarity and accountability.

7. Human-Centered AI

Human-centered AI prioritizes usability, safety, and ethical alignment. It supports humans rather than replaces them, ensuring that AI tools empower users, respect rights, and adapt to diverse needs. This approach is critical for inclusive design and socially responsible innovation.

8. Post-Human AI

Post-human AI introduces a philosophical perspective, challenging the human-centric assumptions of conventional AI. Drawing from posthumanism, it reimagines intelligence beyond the human—considering ecological, Indigenous, and non-Western epistemologies—and calls for pluralistic, culturally grounded approaches to AI development.

9. Algorithmic Bias and Fairness

Algorithmic bias occurs when AI systems perpetuate or amplify social inequalities due to skewed training data or design assumptions. These biases can affect hiring, policing, or healthcare decisions. Fairness in AI requires active efforts to recognize, mitigate, and audit these harms systematically.

10. AI Ethics and Governance

AI ethics and governance involve the policies, principles, and institutional mechanisms guiding responsible AI development. Topics include data privacy, discrimination, transparency, and sustainability. As AI technologies proliferate, global governance frameworks are needed to ensure they benefit all segments of society fairly.

11. Digital governance

The systems, policies, and institutional processes through which digital technologies—such as AI—are regulated, managed, and held accountable. Digital governance involves multiple stakeholders, including governments, private companies, civil society, and international organizations, and shapes how values like fairness, security, and inclusion are implemented in practice.

Foreword

The essence of this report can be summarised in a single sentence: place culture at the center of AI development and application. Only this way can more equity, fairness, and trust be ensured in the international environment.

According to the authors of this report, Saiph Savage and Lili Savage, diversity and inclusivity should be seen as a structural imperative and not as an add-on in AI development. The report sheds light this way on how international engagement in the field of AI regulation and governance should contribute to the safeguarding of Cultural Rights. As current practices use data extraction, continue and amplify existing biases and hierarchies, misrepresent or even omit so-called “data-scarce communities”, the authors call for a shift from “reactive bias mitigation to active harm prevention frameworks” (p. 49), in order to enable “local communities to narrate their [own stories and] experiences for global audiences, promoting cultural visibility beyond borders” (p. 11).

Providing structural tools and enforceable standards for transparency and accountability, as well as engaging at the multilateral level to develop regulations, are also a matter of security policy. Information accuracy, privacy protection, and resilience against cultural domination fueled by geopolitical rivalry are vital domestic security interests, especially to safeguard democracy. As the current state of AI development reflects and enforces global power imbalances, countries should build trusted and sustainable partnerships to mitigate these inequalities, especially by orienting policy design to SDG commitments. As an example for a multilateral AI governance vision, the authors suggest countries disclose cultural impact assessments (similar to environmental ones) in the deployment of AI models and apps.

Various players in international cultural relations (ICR) are very well positioned and skilled in promoting cultural literacy, international collaboration, as well as facilitating partnerships between technologists and cultural actors. As part of the EU’s International Cultural Relations strategy, an investment in cross-regional policy dialogues on these topics should be envisioned in order to make inclusive AI a reality.

This study forms part of ifa's Research Programme "Culture and Foreign Policy", in which experts address relevant issues relating to culture and foreign policy with the aim of involving academics, practitioners, policymakers, and civil society. I would like to thank Saiph Savage and Lili Savage for their excellent work and commitment to this research. In addition, I would like to thank my ifa colleagues Sarah Widmaier and Ivana Putri for their work on the conception, coordination and editing of this project.

International cultural relations are about building bridges, maximizing the benefits of international relations, and mitigating risks through international cooperation. To prevent AI development and application from fostering digital and cultural fragmentation, it is crucial that AI is regulated. This is a matter of global public interest.

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Abstract

Artificial Intelligence (AI) is increasingly embedded in global infrastructures—from governance and education to healthcare and communication—raising urgent concerns about representation, equity, and inclusion. As AI systems are developed and deployed, they often reflect and reinforce dominant cultural norms, marginalizing non-Western languages, epistemologies, and communities. This report explores the systemic risks associated with algorithmic bias, digital colonialism, and cultural homogenization, while also highlighting promising interventions through inclusive design and policy. It presents practical tools—such as cultural impact assessments, fairness-aware auditing, and participatory AI development—to adapt existing systems for diverse contexts. The report calls for strong cross-regional collaboration to ensure AI governance supports cultural sustainability, digital sovereignty, and social justice, placing culture at the core of ethical and inclusive AI futures.

Executive Summary

As Artificial Intelligence (AI) becomes increasingly embedded in global infrastructures—from education and public health to law enforcement and digital communication—AI is transforming how societies function, with profound implications for cultural representation, equity, and inclusion. While much of the ethical AI discourse has focused on technical fairness, transparency, and privacy, this report addresses a critical yet underexplored dimension: the cultural dynamics of AI development and deployment.

AI is not neutral. Current AI systems frequently reflect and reinforce dominant social and cultural norms, often shaped by Western-centric data, assumptions, and values; an outcome of the imbalanced geopolitical concentration of AI development in the Global North¹. This imbalance limits the capacity of the Global South to shape AI in ways that reflect local priorities, values, and epistemologies. As a result, countries in the Global South often function as passive consumers of foreign technologies, reinforcing dependencies and limiting their technological sovereignty.

This leads to the marginalization of underrepresented communities—particularly in the Global South—whose languages, epistemologies, and lived experiences are insufficiently represented in the design and training of AI tools. These systems are rarely designed with cultural pluralism in mind and risk reproducing structural inequities under the guise of objectivity. For instance, facial recognition technologies exhibit higher error rates for individuals with darker skin tones (Buolamwini & Gebru, 2018), while natural language processing (NLP) tools struggle with dialects, idioms, and linguistic expressions from non-Western contexts, resulting in harmful miscommunications and digital exclusion. AI systems used in content moderation often fail to identify context-specific hate speech or discrimination, while disproportionately silencing marginalized voices. Similarly, automated decision-making in hiring,

¹ In this report, the “Global North” is understood as economically and politically dominant regions such as the United States, Europe, and parts of East Asia; while the “Global South” is understood as a blanket term encompassing regions in Latin America, Africa, Southeast Asia, and parts of the Middle East—regions which have historically faced structural inequalities in global economic and technological systems.

healthcare, and policing can exacerbate systemic bias. The report identifies systemic risks such as algorithmic bias, cultural homogenization, and digital colonialism, which are not merely outcomes of technical failures but are rooted in broader epistemic hierarchies and cultural blind spots in global AI governance.

Drawing on interdisciplinary research and stakeholder workshops, this report illustrates both AI's exclusionary harms and the powerful, inclusive interventions being developed in response. A central contribution is its focus on adapting existing systems, not just replacing them. The report offers a practical policy toolkit—including cultural impact assessments, participatory design, and fairness-aware auditing—to embed cultural awareness into the AI lifecycle and move beyond abstract principles. A list of successful initiatives using these methods, from community-led projects to inclusive language technologies, is detailed in the Appendix of this report.

Importantly, this report also addresses the challenge of modifying existing AI systems to make them more inclusive. This includes auditing legacy models, updating datasets, and incorporating community feedback. Just as inclusive methods—such as participatory design and fairness audits—are essential during the development of AI systems, they are equally critical when these systems are already deployed. Institutions using AI in public administration, law enforcement, or digital platforms must recognize that inclusion is not a one-time process but an ongoing responsibility. Retrofitting systems for transparency and fairness requires the same level of cultural sensitivity, stakeholder engagement, and governance as building them from scratch.

Ultimately, the report advocates for a model of inclusive and context-aware AI policymaking that places culture at the heart of ethical governance. It calls for stronger international collaboration—particularly between Western European and Global South actors—to ensure that AI systems support cultural rights, digital sovereignty, and social justice. Examples of such collaboration include efforts to preserve Indigenous languages using AI, support digital education in rural areas, and develop culturally sensitive content moderation tools.

Furthermore, several case studies in this report illustrate both the harms and possibilities of AI in the context of international cultural relations.

Rather than treating inclusivity as an afterthought, the findings position it as a structural imperative. To achieve truly ethical and responsible AI futures, policymakers, developers, and civil society must embrace cultural diversity not only as a value but as a necessary condition for fairness, effectiveness, and long-term trust in AI.

Policy Recommendations and Strategic Outlook

The final section of the report offers concrete recommendations for policymakers, developers, and civil society:

- Embed cultural inclusion into AI policy frameworks at national and international levels.
- Require transparency and accountability in how AI systems are trained, tested, and deployed.
- Fund community-led AI initiatives and research in the Global South to support digital sovereignty.
- Support cross-sector and cross-regional collaboration to foster shared standards of culturally inclusive AI.
- Develop tools and guidelines for assessing cultural impact and adapting existing AI models to new contexts.
- Strengthen participatory governance mechanisms, such as ethics boards and citizen audits, particularly for high-stakes applications, where AI decisions directly impact people's rights, opportunities, or safety (e.g., in law enforcement, border control, hiring, or healthcare).

By aligning AI development with principles of equity, this report envisions a future where technology reinforces—not erodes—our shared human diversity. AI should not only be safe and efficient; it must also be just, inclusive, and capable of serving the full richness of global cultural life.

1. Introduction

Artificial Intelligence (AI) is reshaping the global cultural, economic, and political landscape with unprecedented speed. As AI systems become increasingly integrated into public administration, education, healthcare, creative industries, and international communication, questions of cultural representation, equity, and governance² have become critically urgent. These issues are compounded by growing concerns around safety and abuse, as AI tools are increasingly implicated in online harassment, misinformation, surveillance, and other forms of harm that disproportionately affect marginalized communities. While AI is often framed as a universal tool for innovation and efficiency, its development and deployment are deeply shaped by cultural assumptions, values, and power structures embedded in data, design, and governance.

One of the main reasons AI systems continue to reproduce cultural exclusion is the absence of practical, customizable tools for embedding fairness principles throughout the AI lifecycle. While high-level ethical frameworks—such as UNESCO’s *Recommendation on the Ethics of Artificial Intelligence* and various national AI guidelines—have emphasized the importance of inclusion, there remains widespread uncertainty about who is responsible for ensuring cultural responsiveness in practice (Jobin et al., 2019). This lack of clear accountability results in inconsistent implementation and relegates cultural inclusivity to an afterthought rather than an integral design priority. As a result, AI systems have already caused material harm: from unjust credit scoring and discriminatory hiring practices to misdiagnoses in healthcare and biased policing outcomes (Zowghi & da Rimini, 2023). Feminist scholars argue that addressing these harms cannot rely on technical fixes alone. True algorithmic fairness demands structural reform—greater gender and cultural diversity in AI development teams, transparency in model training processes, and participatory design methods that center the experiences of marginalized communities.

² In this study, “governance” is meant by the policies, regulations, and oversight structures—shaped by governments, private companies, standards bodies, and civil society—that determine who sets the rules for AI development, how decisions are made, and where intervention is possible, from funding and procurement to regulation and auditing.

Despite a growing body of research on AI ethics and algorithmic bias, there remains a significant gap in addressing how AI systems interact with cultural diversity, particularly across underrepresented regions and marginalized communities. Current discourse has largely focused on technical fairness and individual rights, with insufficient attention to the systemic risks posed by cultural homogenization, digital colonialism, and the marginalization of non-Western epistemologies (Noble, 2018). At the same time, global AI governance frameworks often reflect a concentration of influence in technologically dominant regions, leading to asymmetries in how cultural values are encoded, protected, or erased in AI systems.

This report responds to these gaps by exploring how AI both reflects and shapes values, and how it can be governed to support rather than undermine cultural equity. Building on the growing recognition that AI is a cultural as well as technical phenomenon, the report not only investigates the implications of cultural bias in AI systems and of the uneven geopolitical distribution of AI influence, but also looks into concrete technical solutions to enhance AI's security and inclusivity. Drawing from a wide range of case studies, policy analyses, and technical examples, the report highlights both the harms that culturally insensitive AI can cause on marginalized groups, particularly in the Global South, and the opportunities for designing systems that promote cultural diversity. Importantly, the report also examines how existing AI models—many of which are already embedded in high-stakes decision-making environments—can be adapted and improved to better serve diverse cultural contexts and prevent harm. This includes proposing practical interventions such as cultural impact assessments, inclusive data curation practices, participatory design methodologies, and post-deployment auditing mechanisms.

By centering cultural perspectives in AI governance, this report contributes to the emerging field of inclusive AI policy. It recognizes AI's profound influence on digital spaces and social structures, and emphasizes the need for a paradigm shift—one that recognizes the entanglement of technology, power, and cultural meaning, and that supports regional innovation while upholding global ethical standards. In recent years, efforts to create more inclusive AI systems have increased, emphasizing the need to involve stakeholders beyond AI development

teams. Methods like participatory design and user experience (UX) research have been instrumental in identifying biases and integrating diverse perspectives into AI design. Beyond improving product or service functionality, inclusive AI also serves as a means of empowerment and restitution for historically marginalized communities, contributing to fields such as Indigenous AI, feminist human-computer interaction (HCI), and critical race theory (Partnership on AI, 2022).

This report is based on a qualitative research approach that combines literature review, policy analysis, and expert consultation. It also incorporates insights from a stakeholder workshop held in March 2024 to assess challenges and identify actionable strategies for inclusive and secure AI development. The workshop included 56 participants from diverse fields—including policymakers, researchers, AI developers, NGO representatives, artists, and civil society members—from countries such as Mexico, the United States, Germany, France, Sudan, South Korea, Brazil, and Iraq. Discussions focused on AI's potential to create safer online environments, promote gender equality, ensure workplace fairness, decolonize AI for Indigenous empowerment, and advance culturally inclusive policymaking. A thematic analysis was conducted following the event to categorize and refine the insights gathered.

This report is structured as follows: Section 2 introduces the concept of inclusive AI, combining insights from feminist and cultural research with an analysis of geopolitical dynamics and current AI applications. Section 3 explores the security aspects of AI, focusing on issues such as hate speech, gender inequality, algorithmic bias in decision-making, and technical and policy responses. Section 4 offers targeted recommendations for policymakers, international cultural relations (ICR) institutions, and practitioners, outlining strategies for promoting equity, cross-regional collaboration, and culturally aware AI governance. Section 5 concludes the report with a call to action, emphasizing the importance of inclusive and secure AI systems grounded in global cooperation and cultural sensitivity.

2. Inclusive AI

2.1 Overview of Existing (Feminist) Research on Culture, Bias, and AI Development

Artificial Intelligence (AI) systems are deeply embedded in the social, cultural, and political contexts of their development. Feminist and decolonial researchers emphasize that AI is not simply a technical tool, but a socio-technical artifact shaped by the worldviews of its developers (Birhane & Cummins, 2019; Jobin et al., 2019). Contrary to claims of neutrality, AI technologies often reflect the values and assumptions of dominant groups, reinforcing patterns of exclusion and inequality, and resulting in systems ill-equipped to serve culturally diverse populations.

A widely cited example of bias in AI is facial recognition technology, which performs poorly on darker-skinned individuals, especially women. Buolamwini and Gebru (2018) found that commercial facial analysis tools misclassified dark-skinned women up to 35% of the time, compared to less than 1% for lighter-skinned men. These disparities stem from training datasets overwhelmingly composed of lighter-skinned, male faces, and from design processes that overlook inclusion from the outset.

Natural language processing (NLP), a subfield of AI focused on enabling computers to understand, interpret, and generate human language, also reflects entrenched biases. Studies such as that by Bolukbasi et al. (2016) have shown that NLP models often associate male pronouns with professions like “engineer” or “scientist” and female pronouns with roles like “nurse” or “homemaker”. Feminist critiques, such as those elaborated by D’Ignazio et al. (2020), highlight design choices that reinforce stereotypes—for example, using female-coded voices for digital assistants, prioritizing dominant language data in NLP systems, or developing classification models that flatten complex identities. These outcomes are not just technical flaws but manifestations of structural inequality.

Such biases are often embedded from the start. Digital assistants like Siri, Alexa, and Cortana were initially given female-coded voices and subservient personas, reinforcing gendered expectations around care work and compliance (Fosch-

Villaronga et al., 2021). While some companies have since introduced gender-neutral options, these foundational design decisions reflect broader cultural assumptions about gender roles that persist within AI systems.

Cultural hierarchies and linguistic dominance are also reinforced through AI tools like translation engines and content moderation systems. These systems often struggle with non-Western languages, Indigenous dialects, and culturally specific expressions. Moderation algorithms trained primarily in English, for instance, may misclassify activism or culturally significant speech in other languages as harmful content while failing to detect real abuse in underrepresented dialects (UNESCO, 2021).

These harms extend beyond the digital sphere into material impacts. AI-driven hiring tools have been shown to disadvantage women, people of color, and candidates from the Global South by replicating biases present in historical employment data. Amazon's now-abandoned internal tool, for instance, penalized resumes referencing women's colleges or gender-related organizations (Dastin, 2018). Similarly, predictive policing tools and credit-scoring models reflect and perpetuate systemic discrimination by relying on biased historical data (Eubanks, 2018; Obermeyer et al., 2019).

Feminist and decolonial researchers argue that addressing these harms requires more than technical adjustments—it demands structural change. As D'Ignazio et al. (2020) put it, achieving justice in AI “requires rethinking our processes, our teams, our datasets, and our goals from the ground up.” Participatory design—where affected communities are involved from the outset—has emerged as a key strategy. This aligns with feminist Human-Computer Interaction (HCI) principles, which emphasize collaboration, context, and inclusion (Ibid.). Despite their potential, such approaches remain underused in mainstream AI due to time constraints, lack of institutional incentives, or reluctance to share decision-making power.

Another critical issue is epistemic injustice—the marginalization of non-Western knowledge systems. AI-generated search results and recommendation algorithms often privilege Western epistemologies, a phenomenon some scholars

describe as "algorithmic colonialism" (Mignolo, 2011; Noble, 2018). English-language dominance in large language models further narrows the range of cultural perspectives these systems can represent.

Efforts to mitigate bias include fairness-aware algorithms, inclusive datasets, and auditing tools like IBM's AI Fairness 360 and Microsoft's Fairlearn. While helpful, these tools are not a substitute for structural change. Feminist scholars caution that without institutional mandates and meaningful accountability, such measures remain piecemeal. Questions of who defines fairness, who collects and labels the data, and who gets to build the systems must be addressed at every stage of AI development.

Some initiatives demonstrate promising alternatives. Masakhane, a grassroots NLP effort focused on African languages, exemplifies how community-led approaches can support inclusive development (Nekoto et al., 2020). UNESCO's "Recommendation on the Ethics of Artificial Intelligence" (2021) also highlights the need for participatory governance and data sovereignty. However, operationalizing these principles across public and private sectors remains a significant challenge.

Ultimately, feminist and decolonial research not only critiques AI's failures but also reimagines what equitable, inclusive systems could look like. AI can be built to affirm cultural diversity, challenge dominant power structures, and center marginalized voices—but only if justice, not efficiency (typically measured by speed, cost-effectiveness, and predictive accuracy)—becomes the guiding design principle. Inclusion is not a one-time adjustment but an ongoing responsibility that must be embedded throughout the entire AI lifecycle.

2.2 Geopolitics and the Role of Policymakers

Policymakers play a critical role in shaping the development and regulation of AI technologies, but existing governance frameworks often fall short of addressing the cultural dimensions of AI. This remains a widespread issue even as global discourse begins to evolve. For instance, the series of AI Safety Summits

in London (2023), Seoul (2024), and Paris (2025) has progressively worked to broaden the definition of “AI safety” beyond purely technical or existential risks to include principles of fairness and inclusivity. While this high-level dialogue marks a positive step, the primary focus of most implemented national strategies and corporate policies continues to prioritize economic and security interests over the deep, structural integration of equity. While many policy agendas emphasize innovation, competitiveness, and security, they frequently neglect questions of equity, representation, and inclusivity. As a result, AI systems are often deployed without sufficient safeguards to protect the rights, identities, and knowledge systems of marginalized communities—particularly in the Global South.

Most AI policies originate in high-income countries, and their global influence has led to the export of regulatory norms that reflect the priorities of the Global North. Although ethical guidelines like the UNESCO Recommendation on the Ethics of Artificial Intelligence (2021) call for inclusivity and cultural diversity, implementation remains weak. This policy gap has contributed to a form of digital colonialism, where technologies developed in dominant regions are applied elsewhere with little consideration for local contexts.

AI tools used in governance, education, or agriculture, for example, often fail to account for local languages, social structures, or epistemologies. In India, English-based AI systems marginalize speakers of regional languages, reproducing colonial hierarchies (Sambasivan et al., 2021). In Kenya, digital agriculture platforms such as those promoted during the COVID-19 pandemic have been criticized for reinforcing external knowledge systems and overlooking locally embedded farming practices and Indigenous ecological knowledge (Østhassel, 2024). In Brazil, national health platforms such as the RNDS prioritize biomedical data standards while neglecting Afro-Brazilian and Indigenous conceptions of health and care, highlighting a broader misalignment between technological infrastructure and community needs (Massuda et al., 2018). Meanwhile, the development of AI systems frequently relies on the invisible labor of data annotators in the Global South, who are underpaid and excluded from decision-making (Birhane, 2021; Whittaker et al., 2019).

In Southeast Asian countries like Indonesia, Philippines and Cambodia, digital financial platforms that use AI for microcredit scoring have largely been built around Western financial models. These systems often rely on data from formal banking histories, which many rural and Indigenous people do not have. As a result, small business owners—particularly women—are frequently denied access to loans despite their strong participation in informal economies (Zowghi & Bano, 2024).

AI has also failed disabled users in multiple contexts. For example, image recognition algorithms used in education and recruitment platforms have consistently underperformed for users who rely on screen readers or adaptive technologies. Many platforms remain inaccessible due to a lack of multimodal inputs or inclusive UX design. In failing to integrate disability-inclusive datasets and co-design processes, these tools actively exclude disabled populations from digital participation (Whittaker et al., 2019).

In South Africa, predictive policing systems have been criticized for replicating apartheid-era spatial biases. AI algorithms trained on historical crime data have disproportionately targeted Black communities, reinforcing existing patterns of surveillance and mistrust between law enforcement and citizens. These systems often fail to include contextual understanding of social and historical injustices, leading to over-policing in marginalized neighborhoods (Crawford & Calo, 2016).

To address these asymmetries, AI governance must embed cultural inclusion from the outset. This means requiring cultural impact assessments for high-stakes technologies, ensuring Indigenous data sovereignty, and supporting community-led oversight mechanisms. Participatory governance—where communities co-shape the systems that affect them—is essential. Policies must also extend beyond vague ethical commitments and introduce binding mechanisms for transparency, accountability, and redress.

International cooperation is central to this effort, but current models often reinforce global inequities. Global North actors dominate AI policy forums, while stakeholders from the Global South have limited access to influence

decision-making. For instance, regulatory frameworks like the EU AI Act, though ambitious, risk exporting one-size-fits-all standards that overlook the social, economic, and cultural realities of other regions (Csernaton, 2024).

Box 1. A Snapshot of Global AI Regulatory Frameworks

The global approach to AI regulation is fragmented, with major powers adopting distinct models that reflect different priorities. These frameworks have significant implications for how values like fairness, security, and inclusivity are implemented.

The European Union's AI Act³: This is the world's first comprehensive, legally binding framework for AI, establishing a risk-based approach. Unacceptable-risk applications like government-run social scoring and most real-time public biometric surveillance are banned, while high-risk systems (e.g., in hiring, law enforcement, or critical infrastructure) face strict requirements for transparency, human oversight, and accuracy before they can enter the market. The Act aims to set a global standard centered on fundamental rights and user safety.

The United States' Approach⁴: The U.S. has adopted a more sector-specific and non-binding approach, prioritizing innovation and market leadership. Rather than a single law, its governance relies on the presidential Executive Order on Safe, Secure, and Trustworthy AI and voluntary guidelines like the NIST AI Risk Management Framework. This model encourages companies to adopt best practices for managing AI risks but does not broadly prohibit specific applications, leaving regulation to individual sectors like healthcare or finance.

³ For more information see: <https://digital-strategy.ec.europa.eu/en/policies/regulatory-framework-ai> [Accessed on 20.08.2025].

⁴ For more information, see <https://www.nist.gov/itl/ai-risk-management-framework> [Accessed on 20.08.2025].

China's Regulations on AI⁵: China has implemented a state-centric model with specific, binding rules that prioritize social stability and national security. Regulations target generative AI and recommendation algorithms, requiring companies to register their models with the state and ensure content aligns with core socialist values. AI-generated content must be clearly labeled, and practices that undermine state power or create unfair market conditions (like discriminatory pricing) are prohibited, reflecting a top-down approach to governance.

Instead, governance must be co-developed across regions, allowing local expertise to shape global norms. Initiatives such as the African Union's Continental Artificial Intelligence Strategy (2024) and Latin American AI ethics coalitions provide valuable models for collaborative policymaking. Alongside these emerging collaborative efforts, it is important to recognize the influential, state-led AI development model presented by China. Applications like Alibaba's City Brain—which uses AI to analyze vast amounts of public data to optimize urban services like traffic management, enhance urban planning but also raise significant surveillance concerns—reflects a data governance model that is accepted domestically in China, but would provoke privacy concerns in regions like Europe.

This approach represents another distinct, non-Western paradigm in the global landscape, underscoring the geopolitical complexity of establishing shared AI norms. Strengthening these efforts requires redistributing funding, building local capacity, and creating platforms for inclusive policy dialogue.

Policymakers must also regulate the private sector more effectively. Many influential AI systems are built by multinational corporations that often operate beyond the reach of national laws. Without coordinated international standards, these actors can sidestep accountability. Policymakers should mandate public disclosure of training data, independent audits, and human oversight of automated decision-making. Ethics washing—where companies adopt symbolic

⁵ For more information, see: <https://digichina.stanford.edu/> [Accessed on 20.08.2025].

guidelines without enforcement—must be countered with clear, enforceable rules (Bietti, 2020).

Ultimately, fostering inclusive AI requires rethinking governance as a collaborative process grounded in justice and equity. This involves creating space for diverse voices, particularly from underrepresented regions, to shape how AI is built, governed, and used. Only through shared responsibility and meaningful inclusion can AI serve as a force for cultural sustainability and social good.

AI's capabilities in reshaping cultural landscapes globally and its role in promoting cultural understanding are met with both optimism and critical scrutiny by scholars, civil society actors, and international cultural institutions, particularly regarding its capacity to reinforce or challenge existing cultural biases. The cultural limitations of AI are evident in broader power relations and inequalities within international cultural relations. As mentioned in the literature review, most AI systems are developed in the Global North, where they reflect Western values, norms, and biases, and often perpetuate existing power imbalances by favoring characteristics and standards associated with the Global North when exported to the Global South.

On the other hand, AI's potential to foster cultural understanding and inclusivity is particularly evident in efforts to tackle global challenges such as hate speech, gender-based violence, and entrenched power dynamics.

The following sub-section explores the challenges posed by cultural insensitivity in major, globally -deployed AI systems. While it is important to note that many positive, culturally-aware AI initiatives exist that contribute to cultural understanding—a comprehensive list of such projects is available in the Appendix—it is crucial to first examine the systemic biases present in the technologies with the largest global reach. The subsequent examples therefore highlight the risks and shortcomings of some of the most widespread AI applications in use today.

2.3 Existing AI Applications/Systems and Cultural Considerations

Cultural diversity presents both opportunities and challenges for AI systems, especially in addressing bias, linguistic variation, and ethical alignment. Despite their global reach, many AI tools fail to operate equitably across different cultural and linguistic contexts.

ChatGPT, developed by OpenAI, has revolutionized content creation and customer service through human-like responses. However, its training data, primarily sourced from English-language texts, limits its ability to capture idiomatic expressions and cultural nuances beyond Western contexts, thus reinforcing a Western-centric worldview (Floridi & Chirriati, 2023; Karpouzis, 2024).

Google Translate has similarly expanded cross-lingual communication but continues to struggle with low-resource and Indigenous languages. Literal translations often distort meaning in languages such as Guarani or Quechua, and African dialects are frequently misinterpreted. These challenges underscore the need for participatory, community-driven development to preserve cultural and semantic richness (Nekoto et al., 2020; Dueck, 2024; Lewis et al., 2024; O’Heigeartaigh et al., 2020; Ferrara, 2024).

Amazon Mechanical Turk (MTurk) illustrates the ethical tensions of global digital labor. Named after an 18th-century hoax, its branding evokes colonial narratives of hidden labor. Scholars argue that even the name signals ongoing digital exploitation (Gray & Suri, 2019). While MTurk increases access to income, many tasks are shaped by Western norms⁶ and poorly compensated—

⁶ In this context, “Western norms” refer to the cultural and professional assumptions often embedded in task design, which can disadvantage workers from different backgrounds. As detailed further in Section 3.3, these norms often prioritize individualistic communication, Western credentialing systems, and specific language nuances, which can overlook regional differences in qualifications and work structures. Tasks designed with these implicit assumptions may be harder

particularly for workers from the Global South (Shams et al., 2023; Karpouzis, 2024).

Gender bias persists in automated hiring tools. LinkedIn’s algorithms have favored men for high-paying technical roles over equally qualified women (Lambrecht & Tucker, 2019). Broader hiring systems often prioritize Western education or professional experience, marginalizing candidates from the Global South (Mergen et al., 2024; Ferrara, 2024). Amazon’s now-discontinued recruitment tool, which penalized references to women’s colleges, remains a cautionary example (Dastin, 2018).

Voice assistants like Siri have faced criticism for reinforcing gender stereotypes through default female-coded voices. Their voice recognition is also optimized for English—especially American and British accents—leading to misrecognition among users with diverse speech patterns (Roche et al., 2022; Bao & Huang, 2022).

AI’s ethical impact is shaped by cultural context. In China, Alibaba’s City Brain enhances urban planning but raises surveillance concerns. While accepted under China’s data governance model, similar systems would likely provoke privacy debates in Europe (O’Heigeartaigh et al., 2020; Karakaş, 2023).

YouTube’s content moderation algorithms often misclassify advocacy by marginalized groups—flagging feminist or LGBTQIA+ content as inappropriate—while failing to detect context-specific hate speech, especially in non-Western languages (Ferrara, 2024; Matias, 2019).

These failures are especially pronounced in regions where hate speech and gender-based violence take culturally specific forms. In India, caste-based or gendered abuse is often missed by AI systems (Karakaş, 2023). In Africa and Latin America, platform algorithms frequently overlook local abuse while

for workers from the Global South to interpret or complete successfully, potentially leading to unfair rejections and lower compensation.

disproportionately censoring minority voices (Shams et al., 2023; O’Heigartaigh et al., 2020).

Facial recognition technologies have consistently demonstrated racial bias. Amazon’s Rekognition and IBM’s facial analysis tools performed poorly on darker-skinned individuals, leading both companies to restrict law enforcement use. Clearview AI, which scraped billions of images without consent, has misidentified Black and Asian individuals and contributed to wrongful arrests, including one in Detroit (Buolamwini & Gebru, 2018; Birhane & Guest, 2020).

In India, facial recognition used in policing has disproportionately targeted Muslims and lower-caste groups (Birhane & Guest, 2020). In Brazil, similar systems have unfairly flagged Afro-Brazilian individuals, raising concerns about algorithmic racial profiling (Shams et al., 2023; Zowghi & Bano, 2024). Furthermore, predictive policing tools like PredPol replicate systemic bias. In Los Angeles, the system increased surveillance in Black and Latino neighborhoods, reinforcing patterns of over-policing (Shams et al., 2023).

Healthcare AI presents additional risks. VisualDx, a dermatological tool, misdiagnoses conditions in darker-skinned patients due to biased training data. In the U.S., algorithms using cost as a proxy for illness have underestimated Black patients’ needs, limiting their access to care (Obermeyer et al., 2019; Adamson & Smith, 2018).

To address these harms, fairness toolkits like IBM’s AI Fairness 360, Microsoft’s Fairlearn, and Google’s What-If Tool offer metrics and explainability features for bias detection. Yet they remain underutilized—that is, they exist but are rarely integrated into practice due to a lack of standardization and institutional uptake (Richardson & Gilbert, 2021).

Altogether, these examples reveal the urgent need to embed cultural awareness into AI design. While tools like ChatGPT, Google Translate, and City Brain hold transformative potential (see Floridi & Chiriatti, 2020), their limitations expose the dangers of neglecting cultural diversity. Biased AI systems do not just reflect inequality, they help reproduce it. Addressing these gaps requires

inclusive datasets, participatory design, and stronger coordination in global AI governance (O’Heigeartaigh et al., 2020).

To better support inclusive AI development, actors in international cultural relations (ICR)—such as cultural institutions, international NGOs, and bilateral cooperation bodies—can play a transformative role in bridging the gap between AI innovation and cultural equity. These organizations are uniquely positioned to facilitate participatory design by convening local communities, artists, researchers, and civil society groups who are often excluded from AI development processes. For instance, the Goethe-Institut could collaborate with African universities and grassroots language preservation projects to co-create datasets for endangered dialects like Ewe or Shona, supporting fairer natural language processing tools. The Institut français might partner with digital artists and ethicists in francophone West Africa to explore culturally grounded AI ethics through residencies or exhibitions. Similarly, the British Council could support community-led oral history digitization projects in South Asia using AI transcription and archiving technologies.

In addition to content creation and documentation, ICR institutions can advocate for ethical standards that center culture in global AI governance spaces such as the Global Partnership on AI (GPAI) or UNESCO initiatives. They can also offer funding and logistical support for local testing and adaptation of fairness toolkits (e.g., IBM’s AI Fairness 360 or Google’s What-If Tool), ensuring they are interpretable and useful for regional actors. By acting as cultural mediators, these institutions help translate abstract AI ethics frameworks into locally meaningful practices—ensuring that technologies like ChatGPT, Google Translate, or visual recognition systems evolve beyond Western defaults and contribute to genuinely pluralistic digital futures.

3. Advancing Secure AI

3.1 AI to Combat Hate Speech

The proliferation of hate speech on digital platforms has necessitated the development of AI systems designed to moderate online content and protect vulnerable communities. However, one of the biggest challenges in fostering respectful online interactions is bias in content moderation. AI systems struggle to differentiate between hate speech, satire, and culturally specific expressions. The lack of diverse training data exacerbates this issue, making these tools ineffective in detecting hate speech, misogyny, and threats in non-Western languages and dialects, often resulting in unintended consequences such as over-moderation or under-detection of hate speech, particularly against marginalized groups (Matias, 2019). Additionally, algorithmic amplification of harmful content, selective enforcement of moderation policies, and privacy risks for minority language speakers further contribute to the problem. Because minority language content is often underrepresented in datasets, it can be more easily identifiable and less anonymized. This makes users more vulnerable to surveillance, profiling, or unintended data exposure—particularly when AI systems are not designed with adequate safeguards for culturally or linguistically specific communities.

A cultural-specific content moderation example is Jodel in Nigeria, which have deployed AI-driven content moderation systems to tackle hate speech in real-time. These systems are designed to detect and remove abusive content, particularly in local languages such as Hausa and Yoruba. While these tools have been effective in reducing explicit hate speech, they still struggle with more nuanced or coded language that can evade detection (Fosch-Villaronga et al., 2022).

In Brazil, AI-driven content moderation systems have been expanded to tackle xenophobia and racism aimed at Indigenous and Afro-Brazilian populations. Platforms such as X and Instagram have implemented AI algorithms to identify hate speech in Portuguese, but cultural nuances specific to Brazil's diverse population remain difficult for AI to detect. This has led to inconsistent enforcement, with some harmful content remaining online while legitimate

expressions from marginalized groups are flagged in error (Dueck, 2024). Again, AI-driven moderation tools are ineffective at detecting hate speech against Indigenous communities in the Amazon, primarily due to the absence of linguistic data from underrepresented languages. Many Indigenous languages are oral rather than written, meaning AI models trained predominantly on Western textual data fail to recognize harmful discourse in these communities (Lewis et al., 2024).

In Europe, AI has been used to combat hate speech targeting migrant and refugee communities. For instance, platforms like Telegram have introduced AI-driven moderation tools to address hate speech in languages such as Arabic, Dari, and Pashto, which are spoken by many refugees. While these tools have helped reduce explicit hate speech, they often miss cultural subtleties, such as regional dialects and local slang, that are critical to understanding the full scope of the abuse (Drage & Frabetti, 2023).

A similar initiative in India deployed AI-driven moderation systems on platforms such as ShareChat to combat religiously motivated hate speech. These systems focus primarily on detecting explicit hate speech against Muslims and Dalits. However, AI tools have faced significant difficulties in accurately interpreting local slang, dialect variations, and indirect forms of communication often used to evade moderation. As a result, the tools have struggled to consistently identify harmful content, highlighting the complexity of adequately capturing culturally specific linguistic nuances (India Hate Lab, 2025). AI moderation on platforms like WhatsApp struggles to identify hate speech directed at lower-caste communities, as discriminatory language is often encoded in culturally specific terms or regional dialects. This has contributed to unchecked digital violence against Dalits and Adivasis, demonstrating the inadequacy of existing AI models in handling localized forms of discrimination (Birhane & Guest, 2020).

In Myanmar, social media platforms, including Facebook, utilized AI to combat hate speech aimed at the Rohingya minority during and after the crisis in Rakhine State. AI moderation tools were developed to identify and remove content inciting violence against the Rohingya population. Yet, due to limited

Burmese language data and inadequate cultural context understanding, these systems frequently failed to distinguish legitimate political discussions from harmful rhetoric, leading either to the inappropriate censorship of genuine human rights discourse or to unchecked propagation of hate content (Fink, 2018).

Additionally, research focusing on gender-based online harassment in Pakistan highlights the limitations and gaps of current AI moderation systems. Platforms have introduced AI-driven tools to detect and remove misogynistic content in Urdu. Despite these advancements, AI struggles to understand culturally embedded forms of harassment that often utilize sarcasm, local idioms, or euphemisms, leaving significant volumes of harmful discrimination-based online content undetected (Arshad et al., 2023).

Beyond these gaps, AI moderation tools have also been shown to disproportionately censor marginalized voices while allowing dominant cultural narratives to persist unchallenged. In the United States, studies have found that Black activists are more likely to have their posts flagged or removed by AI systems, while hate speech originating from dominant groups often remains undetected (Noble, 2018).

These cases illustrate both the potential and shortcomings of AI-driven content moderation tools. They underscore a recurring gap: AI's limited capacity to adequately comprehend and respond to nuanced and culturally contextualized speech. Furthermore, AI governance structures often lack transparency and public accountability, leaving affected groups with little recourse when AI moderation fails or disproportionately censors their voices. While existing research has acknowledged bias in AI content moderation, there remains a lack of systematic investigation into how these biases manifest across different cultural and linguistic contexts.

This gap is particularly concerning given AI's expanding role in shaping digital governance, online discourse, and global information ecosystems. If AI systems continue to misinterpret or ignore cultural nuances and present selective viewpoints as a result of algorithmic bias, they not only risk amplifying

disinformation, but also perpetuate digital colonialism—a dynamic in which dominant technological paradigms dictate how different cultural groups can engage in online spaces. Large platforms often prioritize profit-driven engagement metrics over inclusivity, meaning that automated moderation tools disproportionately affect minority communities.

To address these issues, further research into culturally informed NLP and moderation strategies is necessary. One solution is to develop AI models that incorporate regional dialects, slang, and cultural expressions. Human oversight should remain a key component of AI moderation systems to counter biases and improve accuracy. Collaboration with linguists, community leaders, and cultural experts can help improve training data diversity. Transparency and accountability must also be prioritized through enhanced reporting policies. Cybersecurity tools can be leveraged to detect manipulation and bias in AI moderation, while crowdsourcing multilingual AI training can ensure linguistic inclusivity and accuracy (Tiribelli et al., 2024).

3.2 AI for Gender Equality

AI systems continue to struggle with gender bias due to historically skewed training data and male-dominated AI development teams. Many AI models fail to accurately represent women across diverse cultural perspectives and reinforce Western-centric gender definitions. Additionally, AI tools have been ineffective in detecting gender-based violence in non-Western contexts, such as misogynistic slurs in Indigenous languages or domestic violence reports in non-English dialects. The UNESCO Women for Ethical AI report (2024) found that voice assistants, recruitment algorithms, and moderation tools often reinforce gender stereotypes, making AI-generated decisions biased against women and gender-diverse groups.

Feminist approaches emphasize the need for AI to better address gender-based violence. For instance, AI-driven content moderation tools have been deployed to detect online harassment, revenge porn, and cyberbullying. However, many of these systems fail to recognize culturally specific forms of abuse—such as

machismo-related harassment in Latin America—resulting in inconsistent enforcement of policies protecting women and gender minorities (Drage & Frabetti, 2023). Context-aware AI moderation models that incorporate feminist epistemologies can help ensure greater accuracy in detecting and mitigating online gender-based violence.

Technology-facilitated gender-based violence (TFGBV), a major form of online gender-based violence (OGBV), is a growing concern, as AI-driven tools are often incapable of recognizing localized patterns of gender discrimination. Deepfake content, AI-generated disinformation, and algorithmic bias in social media contribute to an environment where gendered harassment flourishes without sufficient intervention. AI policy frameworks should incorporate gender-sensitive governance mechanisms to detect and mitigate online violence effectively.

Concretely, these mechanisms include ensuring greater gender and cultural diversity in AI development teams, diversifying training datasets so that women from all backgrounds are well-represented, and establishing AI standards driven by feminist principles to evaluate data quality and algorithmic bias (see Section 4 for further recommendations).

To promote gender equity and inclusivity, AI development must expand and diversify training datasets to ensure women from all backgrounds are well-represented. This includes establishing AI standards driven by feminist principles to evaluate data quality, algorithmic bias, and platform accountability for equity. Furthermore, AI models should be trained to detect gender-based violence across multiple languages and cultural contexts. Transparency in AI-driven hiring and workplace policies must be ensured to prevent discrimination. Encouraging more diverse AI development teams, particularly with increased participation from women and gender-diverse individuals, is crucial. Education programs on AI bias and feminist perspectives should be integrated at all levels of AI development.

3.3 AI in Selection Processes: Addressing Power Imbalances

Advancing ‘secure AI’ extends beyond protecting users from online harassment to ensuring their economic and professional security in high-stakes environments. When AI is used in selection processes—such as hiring, grant applications, or artist residencies—its embedded biases can create profound inequalities by unfairly limiting opportunities for marginalized groups. An AI system that perpetuates discrimination is not secure for those it excludes; it reinforces the very power imbalances that lead to systemic harm. This section explores how biases in AI-driven selection tools threaten professional and economic security and what can be done to address these inequities.

AI-based selection systems are increasingly relevant to cultural organizations and institutions, particularly those with international staff or global outreach through scholarships, grants, or artist residencies. These organizations often operate across linguistic, educational, and cultural boundaries, and are therefore especially vulnerable to the biases embedded in AI-driven recruitment and evaluation tools. Automated systems may inadvertently prioritize Western academic or professional norms, disadvantaging not only applicants from underrepresented regions but also those who did not have the opportunity to acquire formal qualifications. For institutions aiming to support cultural diversity and equity, integrating fair and context-sensitive AI practices is critical to ensuring that global applicants and staff are assessed on equitable terms.

AI in the workplace can reinforce power imbalances by prioritizing Western perspectives, excluding non-Western social norms, and favoring dominant global languages. While many people in the Global South are active users of AI, the report argues that the underlying infrastructure, investment, and policy-making power for AI development are concentrated in the Global North. This forces many countries into consumer roles rather than positions as equal stakeholders in AI creation and governance, further widening global economic and professional inequalities. According to the World Economic Forum (2022), AI-driven recruitment and workplace monitoring tools tend to favor

Westernized professional norms, depriving applicants and employees from underrepresented regions.

Many hiring systems (World Economic Forum, 2022), rooted in Western labor norms—which often prioritize individualistic communication, Western credentialing systems, standardized work hours, and hierarchical management styles—overlook regional differences in communication styles, work structures, and qualifications. Automated screening tools disadvantage non-Western applicants by relying on datasets favoring elite institutions and experiences, amplifying power imbalances in digital workplaces. For instance, AI hiring tools may exclude Global South candidates due to unfamiliar job titles or language nuances, as seen in cases where local roles do not align with pre-trained models.

In online labor marketplaces like Upwork, algorithm-driven team formation prioritizes efficiency over fairness, fostering exclusionary patterns. Research on Fair Team Formation shows how AI unintentionally segregates workforces, favoring certain dominant demographic groups over marginalized groups (Barnabò et al., 2019).

On the other hand, beyond hiring, AI can promote fairness by auditing wage transparency or flagging exploitation and offering governance tools to enforce equity, though institutional inertia often delays action on such insights. AI-driven management systems for evaluations and promotions often operate as opaque “black box” models, raising procedural fairness concerns (Robert et al., 2020). These models lack transparency in how decisions are made, making it difficult for affected individuals to understand, question, or appeal the outcomes. This undermines fairness and accountability, especially in high-stakes decisions related to hiring, promotion, or performance evaluation.

AI systems used for workplace surveillance often disproportionately impact workers from non-Western countries, reinforcing hierarchical labor dynamics between the Global North and South (Birhane, 2021). Surveillance in this context refers not only to efficiency monitoring and skills evaluation, but also to the broader tracking of worker behavior—including productivity metrics, sentiment analysis, keystroke tracking, and even video monitoring. These

systems often operate without transparency or meaningful consent, marginalizing human decision-making in favor of algorithmic assessments and prioritizing productivity over worker rights.

AI can advance Diversity, Equity, and Inclusion (DEI) goals if retrofitted with fairness constraints. Skills-based evaluations, stripped of identifiers like gender or race, reduce bias (Bansal et al., 2022), while AI-driven language analysis can craft inclusive job postings, mitigating exclusionary rhetoric. Yet, reliance on biased historical data—as in Amazon’s gender-biased hiring model (Dastin, 2018)—risks amplifying inequities, underscoring the need for ethical data sourcing and continuous auditing.

Workforce analytics tools can also further equity by identifying wage gaps and promotion disparities, offering governance insights to correct biases (Mujtaba & Mahapatra, 2024). Retrofitting these systems to detect OGBV or discrimination patterns—like harassment in remote work—enhances security, but success hinges on employers acting on findings.

To create a fair and inclusive AI-powered workplace, structural interventions at algorithmic and policy levels are essential. Interdisciplinary collaboration involving sociologists, ethicists, and cultural experts should be encouraged to create unbiased AI models. Integrating fairness constraints, ensuring transparent processes, and embedding human oversight can transform AI into a tool for inclusive workplaces. Introducing human oversight in AI-driven recruitment and performance evaluations will help align decisions with ethical standards.

3.4 Technical Feasibility of Secure AI

Securing AI against OGBV, hate speech, and discrimination is technically feasible but requires significant enhancements to current systems. Retrofitting involves retraining with ethically sourced, diverse datasets, critical for systems like automated hiring, content moderation, and facial recognition, which often perpetuate racial, gender, and cultural biases. This shift rejects algorithmic neutrality, embracing context-aware methods that address structural disparities.

Existing content moderation tools, deployed on platforms like YouTube, falter in detecting OGBV—like revenge porn threats, misogynistic rhetoric in Indigenous dialects or coded hate speech in regional slang—due to limited training data diversity. Enhancing moderation tools requires broader demographic data and bias-auditing techniques to identify and correct discriminatory patterns, leveraging existing infrastructure despite challenges in data access. Advancing NLP by integrating broader linguistic datasets, including low-resource languages, can address this, but ethical sourcing with community collaboration is key to cultural sensitivity and consent (Shrestha and Das, 2023). Additionally, retrofitting NLP to detect intersectional hate speech—e.g., targeting gender and race—requires dynamic, community-informed approaches over static identity models, ensuring marginalized voices are represented despite the complexity of adapting legacy systems.

Techniques like transfer learning, which involve taking a model trained on a large general dataset and fine-tuning it on a smaller, domain-specific dataset, allow pre-trained models to adapt to localized data. This enables detection of implicit threats or context-specific abuse without requiring new infrastructure. For instance, retrofitting NLP to recognize machismo-related harassment prevalent in Latin America could bolster security against OGBV, a gap noted in the workshop discussions conducted as part of this study.

Federated learning further enhances feasibility by training models on decentralized, community-sourced datasets without transferring sensitive data to a central server; instead, local devices collaboratively update a shared model, preserving privacy while tailoring AI to diverse cultural contexts. This could improve hate speech detection in minority languages, reducing the persistent issue of over-censorship of legitimate expressions, and boost trust.

Transparency in data and training processes is crucial to foster accountability (Agarwal et al., 2019). Retrofitting with explainability tools—e.g., clarifying why content is flagged as discriminatory—enhances trust in governance/content moderation applications, allowing users to challenge errors, such as cultural satire flagged as hate speech, thus fostering trust. However, integration into

existing systems requires technical and corporate buy-in often resisted by profit-driven entities.

Mitigating algorithmic amplification of harmful content is equally critical; retrofitting recommendation systems with bias-mitigation layers—e.g., re-ranking to deprioritize violent posts—can curb the spread of OGBV and discrimination, though computational costs and data scarcity for underrepresented groups pose challenges. Co-design with diverse users ensures continuous refinement, but scaling this approach globally demands resources and local expertise often lacking in marginalized regions.

Real-time human-AI collaboration offers a practical retrofit for sensitive governance tasks. Rather than full automation, AI can flag risks—e.g., hate speech in workplace chats—while human moderators, trained in cultural contexts, make final calls, meaning they retain responsibility for the final judgment in complex or sensitive cases. This includes reviewing AI-generated flags, considering cultural and situational context, and overriding or confirming decisions to ensure fairness and accountability, as proposed for legal adjudication and social media governance. Furthermore, human-in-the-loop (HITL) oversight strengthens retrofits, ensuring diverse experts evaluate AI outputs in high-stakes domains like hiring and law enforcement. Interdisciplinary input from ethicists and communities refines models to align with ethical standards (Binns, 2017), as workshop participants emphasized. This hybrid approach mitigates AI's limitations in nuanced judgment, vital for equitable digital spaces—however, coordinating global expertise and funding remains a hurdle, particularly in under-resourced contexts.

Equitable participatory design and data feminism tackle bias at its root, reorienting AI development. Retrofitting via co-design—e.g., involving Indigenous groups to adapt moderation for local hate speech—ensures relevance, aligning with data feminism's call for pluralism and community-led solutions (Cirillo et al., 2020). This shift challenges Western-centric models but risks tokenism without genuine power-sharing.

Ethical data use is another retrofit priority, rejecting extractive practices that exploit communities. Data justice demands trust-based relationships, ensuring transparency in data handling and empowering communities in governance models (D'Ignazio et al., 2020). This means that affected groups should not only be consulted but actively engaged in shaping how data about them is collected, managed, and used. Establishing such trust-based systems is particularly vital in regions with histories of digital exploitation or colonial data practices, where skepticism toward external tech initiatives is high. For instance, retrofitting AI to combat OGBV could involve survivors co-managing datasets, but building such frameworks requires overcoming historical mistrust and logistical barriers in marginalized regions.

Contextual awareness of regional labor practices is critical: retrofitting hiring and candidate selection systems with localized employment data—e.g., recognizing informal economy roles—offers a fairer, context-aware evaluation process. This adaptation, feasible within existing frameworks, counters discrimination albeit sourcing diverse regional data remains a challenge and resistance from cost-focused platforms complicates adoption.

Fairness-aware ranking techniques further enhance inclusivity, as demonstrated by LinkedIn's tripling of representative search results (Geyik et al., 2019). This refers to the increased visibility of underrepresented groups in top search results after applying fairness-aware ranking—a concrete example of how algorithmic adjustments can significantly improve diversity outcomes without sacrificing relevance. Retrofitting hiring platforms with similar re-ranking can boost visibility for underrepresented candidates, countering systemic disadvantages without sacrificing efficiency. This approach could secure AI against discriminatory exclusion and hate speech in workplace interactions, though scaling requires overcoming data and computational constraints in diverse regions. Furthermore, retrofitting AI-driven management systems for evaluations and promotions with explainability tools—e.g., clarifying promotion criteria—can enhance transparency, vital for governance accountability, but integrating these into legacy systems faces technical and corporate resistance.

Despite progress in AI fairness, a comprehensive design agenda remains elusive (Kossow et al., 2021). Isolated fixes such as tweaking datasets fall short without addressing socio-technical structures. Socio-technical structures refer to the intertwined relationship between technology, institutional norms, labor practices, and systemic inequalities that shape how AI is developed, deployed, and governed. For example, even a debiased dataset may be used within an institution that lacks transparency, accountability, or participatory mechanisms. A holistic framework, blending structural transparency, grievance mechanisms, and participatory governance, is needed to shift not just the tools but the underlying systems of power and decision-making. This aligns with workshop participants' calls for systemic change. While coordinating across organizations and regions presents logistical hurdles, several existing initiatives demonstrate that meaningful progress is possible within current systems.

For example, UNESCO's AI for Language Preservation program develops models to document and translate endangered languages spoken by Indigenous communities in Africa, the Amazon, and the Arctic, countering hate speech misidentification due to linguistic bias. Similarly, Google's Woolaroo tool empowers communities to sustain languages like Yugambah in Australia, Māori in New Zealand, and Louisiana Creole by enabling user-contributed data, fostering inclusivity. These efforts demonstrate feasible retrofits for governance applications, yet their reliance on limited funding underscores the need for broader policy support to scale such protections.

To translate these technical possibilities into institutional practice, cultural organizations, funders, and governance bodies should consider concrete structural changes. These include adopting inclusive procurement guidelines that require AI vendors to meet fairness and explainability standards, investing in open-source datasets curated in collaboration with marginalized communities, and establishing participatory oversight boards to audit AI tools used in selection or moderation. Training international staff and grantees in algorithmic literacy can also enhance accountability and empower users to recognize and report biased outcomes. Funding frameworks should prioritize collaborative projects that center underrepresented voices in AI design, such as community-led dataset creation or cross-regional AI ethics research. These

organizational reforms offer cultural institutions a tangible role in shaping inclusive AI governance beyond technical retrofitting alone.

AI reflects its design values. Retrofitting for equity demands technical and governance shifts, a challenge given profit-driven priorities and resource gaps in the Global South. Embedding diverse data, transparency, participatory design, and oversight, this iterative retrofitting process—grounded in collaboration—are first steps towards ensuring that AI serves all equitably, though sustained effort is needed to overcome resistance and resource gaps. Bias audits, fairness-aware algorithms, and human-in-the-loop oversight must be institutionalized to address power imbalances. However, without institutional accountability, technical retrofits alone cannot dismantle structural inequities.

3.5 Retrofitting Existing Technology through Policies

AI technologies offer significant potential to enhance cultural identity and representation, yet initiatives aimed at these goals are often underfunded. As detailed in the Appendix, various inclusive AI applications—particularly those supporting marginalized communities and preserving cultural heritage—have demonstrated real promise when designed or retrofitted with cultural awareness in mind. However, these efforts receive marginal funding compared to commercial or security-focused AI, revealing a clear policy gap across public governance and corporate strategies. Addressing this disparity will require intentional shifts in resource allocation, which remains challenging in an innovation landscape driven by market incentives.

Transparency mandates, such as requiring companies to disclose how AI impacts marginalized groups, can help secure systems against bias and align with calls for greater accountability voiced in stakeholder workshops. Open-source initiatives further democratize development, enabling communities to embed cultural values directly into AI systems. Yet scaling these efforts globally is difficult, due

to corporate resistance and the need for strong policy backing from both government policymakers and corporate decision-makers.

Embedding accountability mechanisms—such as public reporting on AI-driven harms and culturally-sensitive bias audits—can retrofit AI systems for greater inclusivity and safety. These mechanisms must allow affected communities to challenge and correct issues, such as misflagged content that escalates hate speech. Binding international standards are crucial, though hard to implement given fragmented regulatory frameworks and resistance from powerful tech actors.

To advance inclusive governance, institutions involved in AI development and cultural policy—including cultural organizations, tech companies, funding bodies, and civil society groups—must actively engage marginalized stakeholders. Crucially, this engagement cannot occur in silos; progress requires structured, cross-sector collaboration between the technical experts who build AI and the cultural and civil society sectors that understand its real-world impact. This is essential for bridging the significant knowledge gaps that often exist, where tech developers may lack deep cultural context and cultural stakeholders may not be fully aware of AI's technical capabilities and limitations. Identifying who is excluded from decision-making and examining power dynamics between communities and developers can surface structural biases. This knowledge enables more equitable retrofits that address disparities in access, representation, and outcomes. Achieving this demands policy shifts across both public regulation and internal corporate strategy, with a focus on co-creation rather than mere consultation.

Sustained engagement also requires long-term investment. Marginalized communities must be meaningfully involved throughout the AI lifecycle, from design to deployment. This includes co-developing moderation tools for detecting different types of violence occurring online, such as but not limited to online gender-based violence (OGBV), and ensuring flexible, fair compensation for both active participants and those whose cultural content is passively used to train models. Policies that incorporate profit-sharing for data use could address

concerns about digital extraction—though execution remains difficult in under-resourced regions.

Implementing inclusive policies—whether public or private—is an iterative, resource-intensive effort that often stretches smaller organizations. Developing best practices in isolation is inefficient, and policy alone doesn’t ensure compliance. A significant challenge is that staff members within cultural institutions, tech firms, or regulatory bodies may lack the knowledge or motivation to apply ethical standards consistently. This gap often exists because the principles of ‘responsible AI’ are complex and interdisciplinary, while institutional priorities may favor speed and efficiency over deep ethical engagement. Without specific training in areas like algorithmic literacy and clear institutional incentives, even well-intentioned policies can fail to be implemented effectively. Retrofitting governance structures with incentives—such as tax benefits for ethical AI practices—can encourage broader adoption and help build a culture of accountability and inclusion.

3.6 Interconnections Between AI Challenges and Solutions

AI challenges caused by bias—spanning online safety, workplace discrimination, decolonization, and policymaking—are intricately linked. For instance, moderation AI misflagging Indigenous content as hate speech mirrors hiring AI excluding non-Western candidates, both rooted in data scarcity and Western-centric design. Addressing these requires cross-disciplinary collaboration among designers, policymakers, and communities, integrating technical fixes with social and ethical perspectives to retrofit AI for inclusivity across domains.

Retrofitting AI for fairness involves not just diverse datasets but also governance transparency and human oversight, addressing root inequities rather than surface symptoms, a shift requiring interdisciplinary commitment. As Varsha (2023) argues, solutions transcend technical tweaks, necessitating systemic collaboration with ethicists, technologists, and communities. Stakeholder

equity, integrated at every stage—design, deployment, evaluation—is critical, as communities can redefine AI’s purpose and direction (Zdanowska & Taylor, 2022). Having said that, enhancing dataset diversity without rethinking power dynamics risks perpetuating discrimination, as seen in extractive data practices. Cross-team efforts embedding ethics throughout AI’s lifecycle can secure systems against harms, though breaking disciplinary silos remains a persistent hurdle.

Solutions from one area can inform others, enhancing security and equity holistically. Co-design, which is effective in adapting moderation tools to detect OGBV via diverse feedback, can retrofit hiring systems to recognize diverse résumés—e.g., informal roles in the Global South—reducing discrimination. Community-led hate speech detection, successful in low-resource and underrepresented languages, offers a model for recruitment AI, ensuring cultural context informs candidate screening. Similarly, Indigenous governance frameworks prioritizing participatory data collection can guide AI policymaking, amplifying marginalized voices to secure AI against exclusionary biases, though scaling demands resources and trust-building.

AI safety tools designed against harassment—such as content moderation algorithms, automated reporting systems, and support chatbots—illustrate cross-domain lessons for inclusive design. In some cases, chatbots initially failed to identify or respond to gendered violence in Indigenous and non-Western contexts due to cultural mismatches and limited language scope (Drage & Frabetti, 2023). Retrofitting these systems through local co-design—borrowing practices from community-based moderation strategies—enabled more context-aware responses (Lewis et al., 2024).

While some of these adaptations are drawn from real-world pilots and academic case studies, broader implementation remains limited. Nonetheless, these approaches can inform policymaking: emphasizing community-led governance over top-down imposition can strengthen tools to combat online violence and discrimination. Applying similar methods to workplace AI may help address harassment on digital labor platforms, though global application requires navigating logistical and cultural challenges.

Building strong relationships, fostering cross-sectoral engagement, and applying participatory methodologies such as co-design can make AI culturally aware and socially responsible, though this demands incentives and accountability to sustain progress amidst economic pressures.

4. Policy Recommendations

Many ‘responsible AI’ frameworks—including those developed by international organizations, multilateral initiatives, and industry actors—address ethical principles individually rather than as interdependent values. This means they often treat principles like fairness, transparency, and access as separate items on a checklist, failing to recognize that these values are deeply connected and rely on one another to be effective. For example, ensuring equitable access to AI technologies and the opportunities they create must be integrated with principles such as fairness, security, reliability, and transparency (Leslie et al., 2022).

To illustrate this interdependence, providing broad “access” to an AI hiring tool is not a positive outcome if the tool is fundamentally “unfair” and systematically discriminates against certain groups of people. In this case, increasing access only amplifies the harm caused by the lack of fairness. True ethical governance requires understanding that these principles are linked, and failure in one area can undermine progress in all others.

Understanding of ‘responsible AI’ varies widely, even within the same organization—whether a tech firm, public agency, or cultural institution—leading to inconsistent application of inclusive practices. Smaller organizations may lack the resources to develop inclusive policies, which are themselves iterative and complex. Without clear frameworks, institutions must establish best practices independently. Moreover, having a policy in place does not guarantee staff understanding or adherence.

To make progress, policymaking must move beyond aspirational ethics and focus on structural tools, enforceable standards, and partnerships that reflect real-world cultural complexity. The following recommendations propose concrete steps for governments, public institutions, funding bodies, the private sector, as well as institutions and practitioners of international cultural relations to help align AI with values of equity, accountability, and cultural relevance.

4.1 Create Partnerships between the Public Sector, the Private Sector, and Civil Society

Developing culturally inclusive and ethically responsible AI requires multi-stakeholder collaboration, bringing together governments, private sector actors, academia, and civil society organizations. The complexity of AI's societal impact means that no single entity can effectively address issues such as bias, transparency, and cultural representation alone. Through structured partnerships, AI governance can move beyond isolated ethical commitments and become a transformative force for cultural representation and social justice.

- **Collaborate with private sector actors, who drive AI research and deployment.** While governments play a crucial role in setting regulatory frameworks and ensuring that AI deployment aligns with public interests—preventing AI-induced harms while promoting accountability mechanisms (Choung et al., 2023)—regulatory efforts often struggle to keep pace with technological innovation. Private companies must adopt proactive ethical AI development practices, ensuring that their technologies comply with culturally inclusive standards (see Section 4.3. on regulatory mechanisms).
- **Civil society organizations and academic institutions** provide critical oversight and advocacy, bridging the gap between policymakers and the individuals most affected by AI technologies. Community-led AI initiatives have demonstrated the importance of participatory governance, **allowing affected populations to contribute to the design and implementation of AI systems.** By ensuring that cultural groups and historically underrepresented communities are actively involved in AI policy discussions, multi-stakeholder governance structures can help mitigate AI biases and ensure fairer outcomes (Kawakami et al., 2024).
- **Embrace interdisciplinary collaboration in AI development,** incorporating expertise from diverse fields such as anthropology, sociology, and ethics. Cross-disciplinary efforts can ensure that AI systems are evaluated not just for technical robustness but also for their societal implications.

Initiatives such as AI impact assessments and fairness audits should be mandated across industries to provide consistent evaluations of AI systems before deployment.

- **Foster public-private partnerships with AI developers** to facilitate the integration of cultural values into AI systems while leveraging industry expertise in AI ethics and governance (see also Birhane & Cummins, 2019).

Existing AI fairness frameworks and collaborative governance models provide valuable insights into effective multi-stakeholder involvement. Ethical AI toolkits such as IBM's AI Fairness 360 and Microsoft's Fairlearn offer technical guidelines for auditing AI systems, but their implementation depends on industry-wide cooperation and regulatory enforcement (Zowghi & da Rimini, 2023). Collaborative platforms such as the **Global Partnership on AI (GPAI)** and the OECD's **AI Policy Observatory** serve as models for governments, corporations, and advocacy groups to align their AI governance approaches on an international scale.

Ultimately, building an equitable AI ecosystem requires a concerted effort from all stakeholders. **Governments must set clear ethical guidelines, companies must take responsibility for fair AI implementation, and civil society must hold these entities accountable.**

4.1.1 How can the International Cultural Relations sector contribute to these efforts?

In addition to cultural awareness, international cultural relations (ICR) are essential in shaping AI systems that are developed responsibly and inclusively. ICR practitioners foster cross-cultural dialogue and collaboration, helping to embed diverse perspectives into global initiatives. In the AI context, this involves ensuring that technologies reflect social values shaped by different cultural and historical contexts. Cultural relations refers to building and sustaining

meaningful relationships across cultures through dialogue and cooperation—often driven by institutions, artists, scholars, and civil society actors. These relationships foster trust in how technology is developed, governed, and shared across borders. Beyond access, AI must be beneficial to a broad range of communities. Historical experiences with technology influence adoption and trust, requiring responsible frameworks that actively include communities in shaping AI systems to meet their specific needs. Ethical AI development should prioritize partnerships with cultural sectors and marginalized groups, giving them agency in decisions that affect their lives. These collaborations help bridge technological innovation with cultural relevance, resulting in more trusted, effective AI systems.

AI and ICR are deeply intertwined in shaping responsible technology. Stakeholders—including developers, cultural institutions, researchers, civil society, and Global South governments—bring critical perspectives to inclusive AI governance. Initiatives like the UN Global Digital Compact and UNESCO’s Recommendations on the Ethics of AI promote multistakeholder engagement and regional representation. However, without mechanisms for sustained collaboration and power-sharing, such platforms risk falling short. Cultural relations actors must bridge the gap between high-level frameworks and local realities, ensuring AI systems reflect diverse worldviews and are implemented through community-driven processes.

Cultural institutions such as the Goethe-Institut, Institut Français, and the British Council—as well as academics, artists, and community leaders—can help ground ethical AI in local practice. This could include **supporting fellowships for Global South researchers, facilitating cross-regional dialogue between technologists and cultural actors, and co-developing ethical toolkits**. Participatory data stewardship initiatives can further ensure community agency over cultural content. These efforts complement existing frameworks by translating principles into context-sensitive action.

4.2 Integrate Perspectives and Priorities of Diverse Communities in the Multilateral Level

Considering that AI policies and governance frameworks are often shaped by dominant global powers, there is an increasing need for cooperative models that integrate the perspectives and priorities of diverse cultural communities, particularly from the Global South. A key approach is to establish multilateral AI governance structures that actively involve underrepresented regions in decision-making processes, ensuring that AI regulatory frameworks are co-developed rather than imposed by a few leading countries.

4.2.1 Align policies in the Global North and Global South through multilateral AI governance structures:

- **Institutionalise inter-regional policy dialogues to ensure countries in the Global South actively contribute to shaping international AI governance frameworks.** Apart from relying on global AI summits and policy discussions which tend to be led by countries in the Global North, these policy dialogues could be facilitated by dedicated working groups within existing AI policy institutions, such as UNESCO or the OECD, that specifically focus on equitable AI governance (UNESCO, 2021).
- **Establish inter-regional AI policy working groups that facilitate ongoing knowledge exchange.** These collaborative forums should include representatives of countries from the Global South, Indigenous communities, and linguistic minorities, ensuring that AI policies reflect a wider array of cultural contexts. Such initiatives help address power imbalances by fostering equitable contributions to AI ethics, fairness standards, and data-sharing agreements.

- **Work toward the co-development of AI alignment frameworks that recognize regional differences in governance priorities.** Policymakers in the Global South often face unique challenges, such as digital infrastructure limitations, local economic constraints, and historical power asymmetries in global technology governance. Instead of imposing European AI regulatory structures onto the Global South, cross-regional AI working groups should be established to identify shared AI ethics principles while allowing for contextual adaptations. These working groups should include representatives from government agencies, civil society organizations, and technical experts from both the Global North and South, ensuring that AI alignment strategies reflect diverse perspectives (see also Carnegie Endowment for International Peace, 2024). For example, the UNESCO Recommendation on the Ethics of AI (2021) has been adopted in several African and Latin American countries, providing a framework for embedding cultural values into AI development while ensuring international alignment with human rights principles. Encouraging the development of regional AI governance frameworks, such as the African Union’s AI Strategy (2024), can help ensure that cultural preservation remains a key priority in AI policymaking.

- **Establish baseline regulatory standards that prevent exploitative AI deployments.** One of the key issues in AI governance alignment is regulatory arbitrage, where multinational AI firms exploit gaps in AI laws by shifting operations to countries with weaker regulations. Without coordinated global frameworks, AI companies may bypass ethical safeguards in high-regulation regions by deploying AI models in low-regulation areas, exacerbating digital inequalities (George, Duran, & Norris, 2014). International organizations should **work toward a minimum threshold for AI governance**, ensuring that AI developers adhere to ethical principles regardless of where their models are trained or deployed (UNESCO, 2021).

- **Actively support multi-stakeholder AI diplomacy that prioritizes the concerns of the Global South.** This includes establishing structured forums for Global South representatives—be it government officials, researchers, or industry leaders—to influence European AI regulatory frameworks, rather than merely being consulted after policies have been drafted. Bilateral AI

agreements should emphasize knowledge-sharing, capacity-building, and reciprocal investments in AI infrastructure, rather than unilateral impositions of regulatory standards that may not align with local needs (Carnegie Endowment for International Peace, 2024). Moreover, European policymakers, including those working for the EU, should work to remove bureaucratic and financial barriers that prevent Global South nations from meaningfully participating in AI standard-setting bodies such as the European Committee for Standardization (CEN) and the National Institute of Standards and Technology (NIST).

4.2.2 Recognize stakeholders from the Global South as equal, not merely recipients:

- AI governance frameworks must acknowledge the Global South’s **unique expertise in areas such as informal economies, digital inclusion, and grassroots innovation**. Unlike the Global North, where AI regulations often center around corporate accountability and cybersecurity, AI policies in the Global South need to address socioeconomic disparities, linguistic diversity, and community-led AI innovation (Podar et al., 2025). Policymakers based in the Global North should engage with experts from the Global South to integrate lessons learned from successful localized AI initiatives—such as mobile banking platforms in Kenya or AI-driven agricultural solutions in India—into broader global policy frameworks (Podar et al., 2025).
- **Foster South-South cooperation as a parallel strategy to Global North–South engagement** to help amplify Global South priorities in international AI governance. Regional alliances within Africa, Latin America, and the Asia-Pacific should be supported in developing AI policies that reflect localized priorities, which can then serve as a counterweight to dominant Western regulatory frameworks (Carnegie Endowment for International Peace, 2024). By ensuring that AI governance is shaped by a diversity of perspectives rather than a singular Western-centric approach,

diplomatic efforts can foster a more inclusive and responsive global AI ecosystem.

4.3 Enforce Regulatory Mechanisms

Governments play a crucial role in shaping AI's development, deployment, and governance, making it essential that they take proactive steps to ensure AI reflects diverse cultural values. One key strategy is the implementation of national AI strategies that explicitly prioritize cultural inclusivity and diversity in AI research and applications. Many existing AI policies focus heavily on economic growth and technological advancement, often sidelining its societal impact.

To ensure AI governance is culturally responsive, regulatory frameworks must move beyond traditional technical assessments and incorporate context-aware ethical guidelines that reflect the diversity of global AI users. Governments can implement this through interventions elaborated below.

4.3.1 Establish regulatory mandates and participatory governance mechanisms

- To ensure AI systems are culturally responsive, a crucial step is to **require cultural impact assessments** for all high-risk AI systems before their deployment in sensitive domains like education, hiring, healthcare, and law enforcement. A cultural impact assessment is a proactive evaluation—similar to an environmental impact report—designed to identify, predict, and mitigate potential adverse effects on a community's norms, values, traditions, and languages.

This process would involve consulting with local experts and affected groups to analyze how an AI system might reinforce stereotypes, marginalize specific dialects, or misinterpret culturally significant content. The outcome is a detailed report that not only identifies these risks but also outlines concrete mitigation steps, such as dataset enrichment or model adjustments. To be

effective, these assessments must be co-developed with civil society organizations, cultural institutions, and community representatives who can provide essential cultural, linguistic, and social context.

While the European Commission's proposed AI Act is a step in the right direction, its provisions could be significantly strengthened by mandating such cultural impact assessments and inclusivity audits alongside existing ethical compliance requirements.

- Advocate for AI development practices that prioritize the inclusion of local languages, Indigenous knowledge systems, and culturally specific ethical considerations. This can be achieved through regulatory mandates **requiring AI firms to incorporate regionally diverse training datasets and consult local communities** when designing AI applications.
- **Develop advisory boards** with representatives from diverse cultural and linguistic communities to review AI regulation and funding criteria. This also includes **AI ethics boards** that include representatives from historically marginalized communities, cultural scholars, and ethicists. These boards should be granted decision-making power in shaping national AI strategies and investments, overseeing AI regulation and providing guidance on mitigating algorithmic discrimination.
- Facilitate partnerships between AI developers, human rights institutions, cultural organizations, and academia to **co-create AI standards and share case studies of effective community engagement**. AI development must shift from reactive bias mitigation strategies to proactive harm prevention frameworks. One crucial approach is to establish structured accountability mechanisms that center the voices of communities directly impacted by AI systems. This includes the **adoption of community-led auditing practices**, where local organizations, cultural experts, and affected individuals actively participate in reviewing AI models before they are widely deployed. These audits should not only assess bias in training datasets and model outputs but also examine whether the system's objectives align with the needs and rights of the communities it serves (Ciuk et al., 2022).

- Many AI systems rely on extensive data collection, but data sovereignty remains a key concern for developing countries. The dominance of multinational tech firms in AI development has led to data extraction practices that do not benefit the local economies from which data is sourced. To address this, governments should **establish national AI data frameworks that promote fair data-sharing agreements, protect user privacy, and ensure that local communities retain control over their data.**

- Ensure AI provides benefits to underserved communities and reduces inequality. Many AI-driven economic advancements are concentrated in wealthier urban regions, leaving rural and marginalized communities without access to AI-enabled opportunities. To bridge this gap, policymakers should **implement policies that direct AI investment toward public goods,** such as education, healthcare, and infrastructure development. For example, AI-powered diagnostics and telemedicine services can significantly improve healthcare access in remote areas, but only if policies ensure equitable deployment and affordability.

- **Establish a diverse range of incentives for ethical AI adoption:** Beyond punitive measures, regulatory frameworks should include clear incentives to motivate the voluntary adoption of inclusive and secure AI practices. While the report mentions tax benefits, a broader set of motivators can build a more robust culture of accountability. These incentives can include:
 - **Preferential Procurement:** Giving preferential treatment in public procurement processes to AI vendors who can demonstrate adherence to fairness, transparency, and inclusivity standards. This makes ethical practice a significant competitive advantage in securing government contracts.
 - **Access to Dedicated Funding:** Creating dedicated funding streams, grants, or eligibility for digital transformation funds exclusively for cultural and academic institutions that develop participatory AI programs and culturally-grounded innovation.

- **Research and Capacity-Building Partnerships:** Offering access to subsidized training in algorithmic literacy and invitations to join exclusive, publicly-funded research partnerships for organizations that commit to and demonstrate ethical AI governance.
- **Public Recognition:** Creating a government-endorsed public certification or "seal of approval" for AI systems that have successfully passed independent, culturally-sensitive bias audits, providing a strong reputational benefit.

4.3.2 Promote transparency and accountability:

- **Broaden the scope of AI audits to include assessments of cultural impact**—particularly for applications in hiring, education, healthcare, and digital governance to ensure that AI systems are designed and tested with diverse cultural and linguistic inputs. Implementing stricter regulations on data provenance, bias mitigation, and inclusive testing processes will help prevent AI from perpetuating disparities caused by the current training datasets that disproportionately reflect North American- and Eurocentric norms (see Choung et al., 2023).
- Regulatory bodies should **adopt AI fairness auditing standards that assess cultural responsiveness, including biases in AI models**. This requires collaboration between governments, academic institutions, and civil society organizations to create frameworks for evaluating how AI applications impact diverse populations.

Fairness-aware auditing is a specialized review process that goes beyond standard technical performance checks to explicitly test an AI system for social and cultural biases. This involves both quantitative analysis—such as measuring whether the system’s accuracy and error rates are consistent across different demographic groups (e.g., by gender, ethnicity, or linguistic background)—and qualitative analysis, which involves gathering feedback from affected communities to understand the real-world impact of the system’s decisions. Unlike a one-time check, it is an ongoing process of

monitoring and evaluation to ensure the system remains fair as it evolves.

While AI fairness toolkits (e.g. IBM's AI Fairness 360, Microsoft's Fairlearn, Google's What-If Tool) exist and could potentially be applied as a regulatory requirement in AI model evaluation—especially in education, employment, health, and justice sectors—it is important to note that their effectiveness is a subject of ongoing debate. As previously highlighted, these toolkits are often underutilized and are not a substitute for structural change, as they fall short without institutional accountability and mandates for their use. Furthermore, without participatory governance, they risk being applied in ways that do not address the specific cultural contexts and lived experiences of marginalized communities. These tools must be applied alongside qualitative, community-informed evaluation.

- Encourage audits that capture how systems affect marginalized and Indigenous communities in practice by **requiring AI developers to report on culturally responsive AI evaluation metrics**, such as linguistic fairness, regional bias, and ethical data sourcing—not just technical performance metrics. This includes metrics for linguistic inclusivity, algorithmic transparency, and the protection of cultural knowledge systems.
- **Require AI developers to disclose the cultural and linguistic composition of their training datasets.** By embedding these regulatory principles into AI governance, governments can establish safeguards that protect cultural identities while fostering equitable AI adoption across different social and economic contexts.
- **Introduce public-facing accountability dashboards** that track which AI systems are audited for fairness, what methodologies are used, and how community feedback is integrated.

Rather than relying solely on voluntary corporate guidelines, cultural inclusivity must become a matter of public interest and policy oversight. By embedding

these measures into national AI strategies, public procurement rules, and international cooperation frameworks, policymakers can create systemic change.

4.4 Establish Research Partnerships, Funding Programs, and Capacity Building

One of the key obstacles to effective AI collaboration between Europe and the Global South is the misalignment of AI governance models, which assume universal applicability, failing to account for the diverse socio technical realities of Global South nations (Anthony, Sharma, & Noor, 2024). This issue is exacerbated by the fact that countries in the Global South are forced into AI consumer roles rather than being equal stakeholders in AI creation and governance.

To address this, policymakers should proactively work toward creating frameworks for cross-cultural knowledge exchange and AI alignment. This can materialize in the form of research collaborations and capacity building initiatives. As many developing countries lack the resources, expertise, and institutional frameworks needed to regulate AI effectively, capacity-building initiatives are necessary to support AI governance in the Global South.

- The absence of dedicated AI oversight bodies leaves countries in the Global South vulnerable to unregulated deployments that may introduce risks related to privacy violations, algorithmic discrimination, and data exploitation. One potential solution is the **establishment of AI Safety Institutes within countries in the Global South**, modeled after regulatory bodies in the Global North, but adapted to regional needs (see Podar et al., 2025). These institutions could **serve as AI governance hubs**, providing technical expertise, conducting risk assessments, and ensuring AI applications align with local cultural and ethical standards.

- **Create funding streams for culturally grounded AI innovation.** This also includes initiatives centered on cultural and linguistic diversity. Many AI models disproportionately focus on high-resource languages and dominant cultural norms, leaving underrepresented communities without equitable access to AI-driven solutions. By investing in AI projects that document endangered languages, promote Indigenous knowledge systems, and integrate regional epistemologies, governments can help ensure that AI serves as a tool for cultural empowerment rather than exclusion (UNESCO, 2021). Furthermore, supporting academic institutions, artists, and local researchers—especially in the Global South—to lead AI projects rooted in community needs and cultural heritage, as well as making cultural institutions eligible for digital transformation funds when developing participatory AI programs are equally important.
- Support capacity building across sectors by **investing in AI education and training programs that emphasize cultural awareness and interdisciplinary collaboration.** Offer grants and training to public agencies, NGOs, and cultural institutions to build internal understanding of AI ethics, algorithmic literacy, and inclusive procurement. Policymakers should embed this capacity in national digital and education strategies. Public institutions and AI companies alike must be encouraged—if not mandated—to work closely with affected communities, social scientists, ethicists, and community representatives alongside AI development teams in co-designing AI systems that reflect diverse cultural values and traditions (see also Floridi et al., 2018).
- **Emphasize capacity-building initiatives that enhance local expertise in AI governance.** Many countries in the Global South lack sufficient AI research institutions and policy expertise, making it difficult to engage in international regulatory discussions on equal footing. Collaborative programs that support AI education, research funding, and regulatory training in these regions can empower local policymakers to craft AI regulations that are both contextually relevant and globally competitive (Birhane & Cummins, 2019).

- **Prioritize international AI research partnerships to facilitate cross-cultural knowledge exchange.** Joint research initiatives can help build and/or develop AI models that are trained on more diverse datasets, incorporating the linguistic, cultural, and infrastructural specificities of different regions and allowing for culturally attuned algorithmic decision-making. Funding mechanisms, such as dedicated grants for AI research in universities located in the Global South, can help address the resource disparities that often prevent institutions in the Global South from engaging in cutting-edge AI development.
- **Support the development of AI policy toolkits tailored for different Global South contexts.** These toolkits should provide governments with adaptable regulatory templates that allow them to craft AI governance models suited to their specific economic, social, and technological realities. Such an approach ensures that AI alignment is not dictated by a singular Western framework but instead evolves as a result of equitable cross-regional collaboration.

Ultimately, fostering a more balanced AI ecosystem requires policymakers in the Global North to recognize the expertise, agency, and technological aspirations of stakeholders in the Global South, prioritizing collaborative engagement rather than prescriptive regulation.

5. Conclusion

This report has explored the intersections of AI, culture, and policy from a global perspective, revealing the profound implications of cultural considerations in the design, deployment, and governance of artificial intelligence systems. Through workshop insights, literature reviews, and case studies, several key themes have emerged. Among them, the most critical is that culture is not a peripheral concern in AI development—it is central to ensuring equity, fairness, and trust in technological systems.

AI systems are shaped by the cultural assumptions and social values embedded in their training data and design choices. These cultural influences are often implicit and unexamined, which can lead to systems that fail to recognize or respect diverse identities, practices, and languages. As demonstrated throughout the report, when AI systems ignore cultural specificity, they risk reinforcing structural discrimination, perpetuating epistemic injustice, and contributing to the erasure of minority voices.

On the other hand, culturally informed AI design enhances system effectiveness, fosters user trust, and supports more ethical outcomes. Incorporating regional values—such as locally held beliefs about fairness, collective responsibility, social hierarchy, and the role of technology in society—alongside linguistic diversity and local knowledge systems can improve everything from natural language processing accuracy to the fairness of automated decision-making. This requires moving beyond universalist assumptions and embracing culturally adaptive design principles that reflect the pluralistic realities of global societies.

Developing AI systems that respect cultural diversity cannot be achieved in isolation. It demands cross-border collaboration that bridges the technical expertise of developers, the normative frameworks of policymakers, and the lived experiences of local communities. Such collaboration is essential not only to develop inclusive systems but also to amend and improve those already in operation.

Existing AI models, particularly those embedded in governance frameworks or policy applications, often reflect the limitations of their original contexts. Revising these systems to better serve diverse populations requires collective

effort: integrating feedback mechanisms, co-developing localized adaptations, and establishing ethical oversight that reflects regional priorities.

Institutions engaged in international cultural relations are uniquely positioned to support this collaborative work. By investing in education, promoting cultural literacy among AI professionals, and facilitating partnerships between technologists and cultural actors, they can help ensure that future AI systems are both inclusive and contextually grounded.

5.1 Call to Action

To prevent AI from exacerbating existing inequities, all stakeholders must embed cultural diversity as a core value in AI development and regulation. Policymakers must craft inclusive governance frameworks that mandate the cultural evaluation of algorithms, especially those used in critical sectors such as justice, education, and health. This includes updating already-deployed systems to address exclusionary biases and improve contextual relevance.

For AI developers, revising and enhancing current models is as important as building new ones. This requires ongoing audits, responsiveness to community feedback, and implementation of fairness-aware techniques that account for cultural nuance. It also means fostering more diverse development teams and building meaningful partnerships with cultural experts and community stakeholders.

Civil society, including cultural institutions and advocacy groups, plays a pivotal role in holding institutions—such as tech firms, public sector agencies, and international regulatory bodies—accountable and ensuring that AI systems reflect the lived experiences of marginalized populations. These groups must be empowered to help shape both the design and improvement of AI technologies, particularly in regions where digital exclusion is most acute.

Future research should focus not only on building culturally inclusive AI from the ground up but also on refining and transforming the systems already in

operation. This includes developing frameworks for cultural impact assessment, creating inclusive benchmarking standards, and designing modular interventions that allow for the continuous improvement of deployed technologies.

The development of AI policy—spanning national strategies, international standards, and internal corporate governance—must incorporate enforceable mechanisms that ensure AI systems evolve over time in alignment with cultural diversity goals. A broad range of stakeholders, including governments, tech companies, funding bodies, and cultural organizations, should support localized adaptations of global AI models, promote open-source tools for bias detection, and fund community-led efforts to audit and revise AI deployments.

Ultimately, the path toward inclusive AI requires a dual approach: proactive design of culturally sensitive systems and the thoughtful reconfiguration of those already shaping our digital environments. By making adaptability, accountability, and inclusion core principles of AI governance, we can ensure that technological progress benefits all population groups—not just the dominant few.

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Appendix: AI Initiatives around the Globe

Initiative	Region	Technology	Use Case	Challenges	Opportunities	Stakeholders
AI4ALL	Global	Education Programs	Diversity in AI Education	Access Disparities	Empowerment	Educators, NGOs
AI4D Africa	Africa	AI for Agriculture	Farming Prediction	Infrastructure	Food Security	Farmers, Local Gov.
Access Now	Global	AI Advocacy	Digital Rights	Surveillance Risk	Human Rights	Advocacy Orgs
BASAbali	Indonesia	Multimedia Digital Tools	Cultural Preservation	Scalability	Local Empowerment	Community Linguists
Bhashini	India	Translation AI	Language Access	Multilingual Support	Access Equality	Govt of India
CDIAL	Nigeria	Language Digitization	Language Inclusion	Low-resource Language	Digital Preservation	Cultural Centers
Darli	Africa, Asia, South America	AI Chatbot	Regenerative Farming	Cross-cultural Relevance	Localized Advice	Farmers, NGOs
Data Science for Social Good	United States	AI Tools for Social Good	Youth Outreach	Impact Evaluation	Social Impact	Researchers, NGOs
Dunhuang Cave Preservation	China	AI Restoration	Art Preservation	Ethical Design	Digital Access	Art Historians
FUNREDES	Dominican Republic	Open Source Platforms	Language Empowerment	Sustainability	Linguistic Justice	Civil Society
Google Quechua Integration	Peru	Translation AI	Language Preservation	Translation Accuracy	Inclusion	Google, Gov.
India National	India	Museum Digitization	Heritage Access	Data Ownership	Virtual Access	Govt of India

Initiative	Region	Technology	Use Case	Challenges	Opportunities	Stakeholders
Digital Repository						
Indigenius Mobile	Nigeria	Conversational AI	Multilingual Communication	Infrastructure	Linguistic Inclusion	CDIAL
Khon Dance Digitization	Thailand	Movement Recognition AI	Dance Heritage	Data Labeling	Cultural Education	Thai Ministry
Laboratoria	Latin America	Training Programs	Women in Tech	Funding	Gender Equity	Women Orgs
La Casa Ida	Peru	Media Lab	Digital Arts	Defunding	Experimental Culture	Artists
LatamGPT	Latin America	Language Model	Inclusive Language AI	Dataset Gaps	Multicultural AI	Govt of Chile
LatinX in AI	Latin America	Mentorship and Advocacy	Latinx Representation	Visibility	Leadership Diversity	Latinx AI Community
Masakhane Project	Africa	NLP Tools	African Language Inclusion	Scalability	Language Access	Local Developers
Mexico Yandex Translate Adaptation	Mexico	Translation Platform	Indigenous Language	Dialectal Accuracy	Cultural Inclusion	NGOs, Gov.
Microsoft Mexico Investment	Mexico	Cloud Infrastructure	SMB Support	Equity in Access	Economic Growth	Microsoft
National AI Policy (Colombia)	Colombia	Policy	Research & Ethics	Funding Gaps	Governance Ethics	Colombian Govt

Initiative	Region	Technology	Use Case	Challenges	Opportunities	Stakeholders
Peru Language Tools	Peru	Translation, Education Tools	Language Revitalization	Tech Alignment	Cultural Identity	Peruvian Linguists
Runasimipi Qespisqa	Peru	Open-source Spellcheck	Spellchecking Tools	Technical Barriers	Linguistic Tools	Peruvian Developers
Sahabat AI	Indonesia	Language Model	Local Language AI	Scalability	Inclusive AI	Sahabat AI Team
Silano Oral Histories	Canada	Transcription AI	Oral History Archiving	Bias	Cultural Continuity	First Nations
TAIK	Africa	Storytelling Platform	STEM Empowerment	Narrative Control	Digital Storytelling	TAIK Foundation
Thailand Khon AI	Thailand	Preservation AI	UNESCO Heritage	Pose Recognition	Art Conservation	Thai Gov.
Time Machine Project	Europe	Digitization AI	Historical Archives	Taxonomy Misuse	Knowledge Preservation	EU Cultural Bodies
Tupuy App	Peru	Voice Recognition	Learning Tool	Speech Limitations	Education	Language Learners
UNESCO Guidelines	Global	Policy Guidelines	Cultural Diversity	Implementation	Framework Development	UNESCO
VisualDx	USA	Dermatological AI	Skin Diagnosis	Bias in Training	Health Equity	Health Providers
World Meteorological Organization (DRR)	Global South	Disaster Risk AI	Climate Warnings	Infrastructure	Resilience Building	WMO, Gov. Agencies
Zowghi & Bano LXAI	Latin America	Advocacy Coalition	Equity Promotion	Funding Gaps	Research Collaboration	AI Ethics Orgs

Initiative	Region	Technology	Use Case	Challenges	Opportunities	Stakeholders
Zowghi & Bano AI4ALL	Global	Education Outreach	Education Access	Sustainability	Mentorship	Educators
Amazon Rekognition	USA	Facial Recognition	Bias Review	Racial Disparity	Bias Accountability	Amazon
Clearview AI	USA	Facial Recognition	Surveillance Bias	Misidentification	Ethical Review	Clearview AI
City Brain	China	Urban Planning AI	Data Governance	Governance Ethics	Policy Innovation	Chinese Gov.
IBM Fairness Toolkit	Global	Bias Detection Tools	Toolkit Audits	Adoption Rates	Bias Mitigation	Developers, Auditors
LinkedIn Algorithm	USA	Hiring Algorithm	Gender Hiring Bias	Bias Metrics	Bias Repair	Recruiters, HR

About the Authors

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Imprint

This study is created within the framework of ifa's Research Programme "Culture and Foreign Policy" and is published in the ifa Edition Culture and Foreign Policy series.

The Research Programme is funded by the Federal Foreign Office.

The views expressed are those of the author and do not necessarily reflect those of ifa.

Publisher:

ifa (Institut für Auslandsbeziehungen e. V.),
Charlottenplatz 17,
70173 Stuttgart,
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info@ifa.de,
www.ifa.de
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Authors:

Saiph Savage, Lili Savage

Editing:

ifa's Research Programme "Culture and Foreign Policy"

Cover Photo Credits:

Unsplash/Google DeepMind – An artist's illustration of artificial intelligence (AI). This image represents ethics research understanding human involvement in data labelling. It was created by Ariel Lu as part of the Visualising AI Programme, an initiative by Google DeepMind.

Design:

Andreas Mayer, Stuttgart

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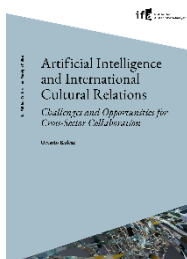


DOI: <https://doi.org/10.17901/1570>

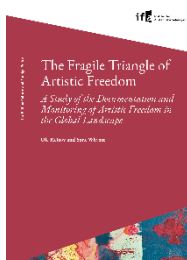
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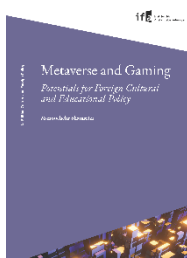
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Inclusive and Secure Artificial Intelligence

A Global Perspective on Policy and Technical Developments

Artificial Intelligence (AI) is increasingly embedded in global infrastructures—from governance and education to healthcare and communication—raising urgent concerns about representation, equity, and inclusion. As AI systems are developed and deployed, they often reflect and reinforce dominant cultural norms, marginalizing non-Western languages, epistemologies, and communities. This report explores the systemic risks associated with algorithmic bias, digital colonialism, and cultural homogenization, while also highlighting promising interventions through inclusive design and policy. It presents practical tools—such as cultural impact assessments, fairness-aware auditing, and participatory AI development—to adapt existing systems for diverse contexts. The report calls for strong cross-regional collaboration to ensure AI governance supports cultural sustainability, digital sovereignty, and social justice, placing culture at the core of ethical and inclusive AI futures.